Topic: Euler's method

Question: If $y(1) = 0$, use Euler's method to approximate $y(3)$ with $n = 4$ steps.

$$y' = \frac{1}{2 + y}$$

Answer choices:

A $y(3) \approx 1$

B $y(3) \approx 1.69$

C $y(3) \approx 0.86$

D $y(3) \approx 0.82$

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Solution: C

We're starting at $y(1)$ and we need to get to $y(3)$ in $n = 4$ steps, so

$$\Delta t = \frac{3-1}{4}$$

$$\Delta t = \frac{1}{2}$$

Starting with $(t_0, y_0) = (1,0)$, and with $\Delta t = 1/2$, we build our table and use it to approximate $y(3)$.

$$t_0 = 1 \qquad y_0 = 0 \qquad y_0 = 0$$

$$t_1 = \frac{3}{2} \qquad y_1 = 0 + \left(\frac{1}{2+0}\right)\left(\frac{1}{2}\right) \qquad y_1 = \frac{1}{4}$$

$$t_2 = 2 \qquad y_2 = \frac{1}{4} + \left(\frac{1}{2+\frac{1}{4}}\right)\left(\frac{1}{2}\right) \qquad y_2 = \frac{17}{36}$$

$$t_3 = \frac{5}{2} \qquad y_3 = \frac{17}{36} + \left(\frac{1}{2+\frac{17}{36}}\right)\left(\frac{1}{2}\right) \qquad y_3 = \frac{2{,}161}{3{,}204}$$

$$t_4 = 3 \qquad y_4 = \frac{2{,}161}{3{,}204} + \left(\frac{1}{2+\frac{2{,}161}{3{,}204}}\right)\left(\frac{1}{2}\right) \qquad y_4 \approx 0.86$$

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Topic: Euler's method

Question: If $y(0) = 0$, use Euler's method to approximate $y(4)$ with $n = 5$ steps.

$$y' = 1 + y$$

Answer choices:

A $y(4) \approx 8.50$

B $y(4) \approx 17.90$

C $y(4) \approx 55.60$

D $y(4) \approx 9.50$

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Solution: B

We're starting at $y(0)$ and we need to get to $y(4)$ in $n = 5$ steps, so

$$\Delta t = \frac{4-0}{5}$$

$$\Delta t = \frac{4}{5}$$

Starting with $(t_0, y_0) = (0,0)$, and with $\Delta t = 4/5$, we build our table and use it to approximate $y(4)$.

$t_0 = 0$	$y_0 = 0$	$y_0 = 0$
$t_1 = \frac{4}{5}$	$y_1 = 0 + (1+0)\left(\frac{4}{5}\right)$	$y_1 = \frac{4}{5}$
$t_2 = \frac{8}{5}$	$y_2 = \frac{4}{5} + \left(1+\frac{4}{5}\right)\left(\frac{4}{5}\right)$	$y_2 = \frac{56}{25}$
$t_3 = \frac{12}{5}$	$y_3 = \frac{56}{25} + \left(1+\frac{56}{25}\right)\left(\frac{4}{5}\right)$	$y_3 = \frac{604}{125}$
$t_4 = \frac{16}{5}$	$y_4 = \frac{604}{125} + \left(1+\frac{604}{125}\right)\left(\frac{4}{5}\right)$	$y_4 = \frac{5{,}936}{625}$
$t_5 = 4$	$y_5 = \frac{5{,}936}{625} + \left(1+\frac{5{,}936}{625}\right)\left(\frac{4}{5}\right)$	$y_5 = \frac{55{,}924}{3{,}125}$

$$y_5 \approx 17.90$$

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Topic: Euler's method

Question: Use Euler's Method with step size 0.5 to approximate the values y_1, y_2, y_3, and y_4 when $y(3) = 0$, then find $y(5)$.

$$y' = y - 2t$$

Answer choices:

A $y(5) \approx -28.5$

B $y(5) \approx -16$

C $y(5) \approx -47.75$

D $y(5) \approx -16.5$

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Solution: A

Since we're given step-size directly, we already know that

$$\Delta t = 0.5$$

To start building our table, we identify $t_0 = 3$ and $y_0 = 0$ from our initial condition, $y(3) = 0$. Since we need to find y_4, we'll start adding $\Delta t = 0.5$ to t_0 until we reach t_4.

$$t_0 = 3 \qquad y_0 = 0 \qquad\qquad y_0 = 0$$

$$t_1 = 3.5$$

$$t_2 = 4$$

$$t_3 = 4.5$$

$$t_4 = 5$$

Now that we've built the outline for our table, we'll use Euler's formula to find y_1, y_2, y_3, and y_4.

$$t_0 = 3 \qquad y_0 = 0 \qquad\qquad y_0 = 0$$

$$t_1 = 3.5 \qquad y_1 = 0 + (0 - 2(3))(0.5) \qquad y_1 = -3$$

$$t_2 = 4 \qquad y_2 = -3 + (-3 - 2(3.5))(0.5) \qquad y_2 = -8$$

$$t_3 = 4.5 \qquad y_3 = -8 + (-8 - 2(4))(0.5) \qquad y_3 = -16$$

$$t_4 = 5 \qquad y_4 = -16 + (-16 - 2(4.5))(0.5) \qquad y_4 = -28.5$$

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After filling out the table, we can say that the value of $y(5)$ is approximately

$$y(5) \approx -28.5$$

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